

FieldForce AI

Product Design / Product Requirement Document (PDR / PRD)

Pharma + FMCG Field Force Management Platform | Mobile App + Admin Panel + Cloud Backend

1. Product Vision

FieldForce AI is a mobile-first field force management ecosystem for Pharma and FMCG sales teams. It manages attendance, GPS tracking, beat plans, outlet visits, order booking, sales reporting, schemes, targets, incentives, gamification, product promotions, FMCG retail audits, communication, and AI-assisted field execution.

- Primary users: MR, ASM, RM, telecalling team, FMCG sales team, distributors, stockists, admin, and management.
- Current MVP is suitable for pilot testing with 2-3 field users.
- Long-term direction is a flexible FieldAssist/Bizom/BeatRoute style operating system with stronger AI and gamification.

2. Current MVP Modules

Mobile Attendance: GPS check-in/check-out base workflow and cloud attendance save.

Outlet Session: start outlet check-in, check-out, time at outlet, KM entry, cloud report.

Visit Reporting: customer selection, outcome, notes, and DCR cloud sync.

Order Booking: customer dropdown, product search, quantity, order total, cloud save.

Customer Master: admin/mobile customer add, search, cloud listing, CSV import.

Product Master: admin create/import; mobile read-only product details and order usage.

Beat Plans, Schemes, Reports, Targets, Promotions, FMCG Audit, Communication and CSV Import foundation are added.

3. Field User Workflow

1. Open mobile app and login with user name and role.
2. Mark GPS Check In to start field day.
3. Open Beat Plan and review planned customers/outlets/doctors.
4. Select customer in Visit screen and tap Start Outlet Check-In.
5. Enter KM from last outlet and tap Check Out Outlet.
6. Submit Visit Report, Book Order, submit FMCG retail audit where required, then end day with GPS Check Out.

4. Admin Workflow

7. Create or import customer master and product master.
8. Create territory/state/area mapping and transfer areas when needed.
9. Create beat plans and assign them to field users.
10. Create schemes, discounts, and product promotion campaigns.
11. Set monthly, product, team, and territory targets.
12. Publish announcements and training messages.
13. Review visits, orders, attendance, outlet time, KM, retail audits, and performance reports.

5. Bulk Data Import Procedure

Customers CSV columns: type, name, mobile, state, area, customer_class, specialty, outstanding. Required field: name.

Products CSV columns: name, composition, category, pack, mrp, sale_rate, scheme, stock. Required field: name.

Areas CSV columns: state, city, area, territory, assigned_manager. Required field: area.

Beat Plans CSV columns: title, assigned_to, area, customer, planned_date, sequence_no. Required field: title.

Tasks CSV columns: title, assigned_to, priority, due_date. Schemes CSV columns: title, scheme_type, product, customer_segment, rule_text.

14. Open Admin Panel and go to the required module.
15. Click Sample CSV and download the format.
16. Fill your data in the same columns without renaming headers.
17. Click Import CSV and upload the completed CSV file.
18. After import, the module refreshes automatically. Mobile app will show synced data after refresh/open.

6. Recommended Scalable Architecture

Mobile App: current PWA; future React Native or Flutter native app.

Admin Panel: current static web app; future Next.js management portal.

Backend: Supabase REST now; future Node/NestJS service layer for enterprise APIs.

Database: PostgreSQL/Supabase with RLS, audit logs, multi-company support.

Realtime: Supabase Realtime/WebSockets for live tracking and notifications.

Notifications: FCM, WhatsApp API, SMS gateway.

AI: OpenAI APIs, OCR, prediction models for route suggestions, sales assistant, stock recommendations.

7. Database Modules

- Current: customers, products, visits, orders, attendance, outlet_sessions.
- Operations: beat_plans, tasks, schemes, territories.
- Enterprise foundation: targets, promotions, retail_audits, announcements, expenses, collections.
- Future: companies, users, roles, permissions, audit_logs, invoices, payments, rewards, incentives, location_points.

8. Roadmap

Pilot: 2-3 MR testing, product/customer import, beat plan, attendance, visits, orders, outlet time and KM tracking.

Operations: role-based login, approvals, expense/collection workflows, returns, invoice visibility, distributor portal.

Scale: multi-company, hierarchy, audit logs, background GPS, fake GPS detection, offline sync engine.

AI: predictive sales, missed visit prediction, route optimization, OCR bill scanning, smart stock and target suggestions.

Enterprise: native apps, high-concurrency APIs, reporting warehouse, ERP/Zoho/WhatsApp/payment integrations.

9. Immediate Testing Checklist

- Run latest Supabase schema SQL after each database update.
- Push latest GitHub commits so Vercel updates the live link.
- Import products and customers using Sample CSV and Import CSV.
- Open mobile link on field user phone and confirm product/customer data appears.
- Test GPS Check In, Start Outlet Check-In, Check Out Outlet with KM, Submit Visit, Book Order.
- Open Admin Reports and confirm outlet time, KM, visits, orders, and imported master data show correctly.

10. Live Links

Mobile App: <https://fieldforce-ai.vercel.app/mobile>

Admin Panel: <https://fieldforce-ai.vercel.app/admin>

GitHub Repository: [siddharthsingl25-del/fieldforce-ai](https://github.com/siddharthsingl25-del/fieldforce-ai)

Cloud Database: Supabase project connected through publishable API key for MVP testing.

Important Note

Current build is an MVP suitable for pilot testing. Before large-scale rollout, add role-based authentication, private RLS policies, user hierarchy, audit logs, secured API layer, background location handling, and offline sync.